

Sri Lakshmi Annapurna Tiffin Center

Complete Web Application System & Architecture Documentation

Application Name:	Sri Lakshmi Annapurna Tiffin Center	Version:	v1.0.0 (Production)
Live Website (Render):	https://annapurna-tiffin.onrender.com	Local Dev Server:	http://localhost:3000
GitHub Repository:	aravindkasarla5/annapurna-tiffin	Primary Stack:	Node.js, Express.js, SPA JS, Vanilla CSS

1. System Overview & Key Modules

Sri Lakshmi Annapurna Tiffin Center is a modern full-stack web application designed for online food ordering, live order tracking, payment verification, owner management, customer rewards, and review analytics. The system provides dedicated portals for both **Customers** and **Hotel Owners**.

Core Customer Portal Features:

- **Authentic Tiffin Ordering:** Interactive menu categorized by Breakfast, Lunch, Dinner, Snacks, and Daily Specials.
- **Live Order Status Tracking:** Real-time status sync (Received → Preparing → Ready → Completed → Rejected).
- **Interactive Payment Gateway:** Supports Cash on Delivery / Dine-in and Online UPI payments with shop owner QR scanner, UTR transaction verification, and manual receipt uploads.
- **Referral Rewards Program:** ■30 first-order referral discount applied automatically via unique referral links or codes.
- **30-Minute Inactivity Session Timeout:** Automatic client & server-side session expiration after 30 minutes of inactivity while preserving account history and past orders.
- **Customer Support & Live Chat:** Integrated ticket creation and real-time support messaging.

Core Hotel Owner Portal Features:

- **Real-Time Order Management Hub:** Filter orders by status, update preparation state, approve UPI payments, and print customer bills.
- **Live Tiffin Menu & Availability Control:** Instant ■ Open / ■ Closed store status toggle and item-by-item availability switches.
- **Customer Reviews & Ratings Hub:** KPI analytics (Average Star Rating, Total Reviews, Rating Distribution Bars), official owner response modal, and public visibility toggle.
- **UPI Payment & QR Settings:** Upload custom store UPI QR code image and specify UPI ID for instant customer scanning.
- **Analytics & Revenue Reports:** Daily sales summaries, top-selling tiffins breakdown, and payment method statistics.

2. Technical Architecture & Security

The application follows a clean Single Page Application (SPA) architecture on the client side coupled with a lightweight RESTful Express.js backend server.

Layer	Technology & Implementation
Backend Server	Node.js (v18+) & Express.js REST API with CORS, Bearer Token Auth, and JSON storage.
Database	Local JSON Database (db.json) managed via atomic file-system read/write operations (loadDB & saveDB).
Frontend UI	Vanilla JavaScript (app.js SPA controller), HTML5 semantic structure, FontAwesome 6 icons.
Styling System	Custom Vanilla CSS (styles.css) featuring modern dark glassmorphic design, HSL colors, responsive media queries across desktop, tablet, and mobile (320px–1200px+).
Session Security	Bearer token tracking in db.tokens with server-side last_activity timestamping. Enforces 30-minute inactivity expiration for Customer accounts while excluding background status polling (X-Background-Poll).

3. Customer Session Expiration Logic

To ensure robust user security without interrupting active browsing, session expiration follows a strict rule set:

- Customer Timeout:** Set to exactly 30 minutes (1,800,000 ms) of continuous inactivity.
- User Interaction Reset:** Mouse movements, clicks, keyboard typing, touch events, and navigation reset the timer.
- Background Polling Exemption:** Automatic 2-second background status updates send X-Background-Poll: true headers and do *NOT* reset the token activity timer on client or server.
- Server Verification:** Protected endpoints verify `Date.now() - last_activity <= 30 minutes`. If expired, server returns `HTTP 401 { expired: true }`.
- Data Integrity:** Session expiration clears only client cache (localStorage/sessionStorage). Customer orders, wallet balance, and profile remain 100% safe in db.json.

4. API Endpoints Reference

Method	Endpoint Path	Auth / Access	Description
POST	/api/auth/register	Public	Register new Customer account.
POST	/api/auth/login	Public	Authenticate User / Owner & return Bearer token.
GET	/api/menu	Public / Optional Auth	Fetch all tiffin items & availability status.
POST	/api/orders	Customer Auth	Submit new tiffin order with payment details.
GET	/api/orders	Authenticated User	Get customer orders or all hotel orders (Owner).
PATCH	/api/orders/:id/status	Owner Auth	Update order status (Preparing/Ready/Completed).
GET	/api/reviews	Public / Owner Auth	Fetch reviews & owner rating statistics.
POST	/api/reviews/:id/reply	Owner Auth	Post official hotel owner response to customer review.
GET	/api/support	Authenticated User	Fetch support tickets and live messaging history.

5. Default Credentials & Access Information

Role	Mobile Identifier	Password	Permissions & Scope
Hotel Owner / Admin	9392874900	Kumar@9392	Full store management, menu edit, orders hub, review replies, revenue reports.
Customer Account	9876543210	Kumar@9392	Browse menu, add to cart, checkout, live order tracking, wallet redemption, review submission.

6. Deployment & Maintenance Summary

- **Local Environment Execution:** Run npm start inside project folder. Server listens on port 3000.
- **Production Cloud Deployment:** Live on Render.com automatically connected to GitHub main branch.
- **Live Website URL:** <https://annapurna-tiffin.onrender.com>

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